FIVUCSAS

Face and Identity Verification Using Cloud-based SaaS

Multi-tenant biometric authentication with active-liveness anti-spoofing

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Abstract. *Consumer authentication remains fragmented across leak-prone passwords, SIM-swappable OTPs, and biometric services that trust a* single passive *signal a prepared attacker can replay. FIVUCSAS proposes a tenant-configurable, multi-modal identity platform where any subset of* ten *authentication factors composes into an MFA flow and every biometric factor is wrapped in a randomised* active *liveness challenge. The system is decomposed along its right axis — a frozen OIDC contract on one side, an iterating ML pipeline on the other — and runs end-to-end face verification in under one second on a single commodity CPU server. We report 0 / 120 replay-attack false-accepts and 0.97 passive-spoof AUC on an internal benchmark, with the full deployment surface and 1 820+ tests gating every change.*

I. Problem statement

Online authentication is fragmented. Passwords leak, SMS OTPs are SIM-swappable, hardware keys exclude phone-only users, and most biometric services trust a single passive signal — a still selfie or a recorded clip — that a prepared attacker can replay. Meanwhile GDPR and KVKK demand verifiable consent, the right to erasure, and explicit data portability; consumer authentication stacks treat these as an afterthought.

Goal. A tenant-configurable, multi-modal identity platform where any subset of ten authentication factors composes into MFA flows, and where every biometric factor is protected by an *active*, randomised liveness challenge a recorded video cannot replay.

II. Contributions

This work makes the following concrete contributions to the practice of multi-tenant biometric identity verification on commodity hardware:

C1. A **tenant-configurable composition** of ten authentication factors (PASSWORD, EMAIL_OTP, SMS_OTP, TOTP, QR, FACE, VOICE, FINGERPRINT, WEBAUTHN, NFC_DOCUMENT) where each tenant declares the set of valid MFA flows at the admin layer.

C2. An **active randomised liveness** protocol — per attempt the server emits an n-step facial-action sequence (n=3–5) bound to a nonce; the client returns a 60-frame landmark stream; the server verifies each action against EAR / MAR / yaw and temporal consistency, in parallel with a UniFace MiniFASNet passive spoof check.

C3. **Pipeline decomposition** along the OIDC-contract / ML-iteration boundary into a Spring Boot identity-core and a FastAPI biometric-processor, isolated on a Docker network and gated by an internal API key.

C4. **Embedding-at-rest encryption**: Facenet512 embeddings are wrapped in Fernet (AES-128-CBC + HMAC-SHA-256) prior to write into pgvector; the HNSW index search path operates on plain vectors only inside the trusted process.

C5. A **regulator-aware data lifecycle**: tenant-scoped soft-delete, scheduled purge, and audit-logging satisfying GDPR Art. 15 / 17 and KVKK equivalents.

III. Passive vs. active liveness

Passive only. Single frame → texture / colour cues.
- ✗ Photo of phone screen *often passes*
- ✗ Recorded clip *passes*
- ✗ Deepfake injection *often passes*

Active challenge. Randomised n-step facial-action sequence + landmark check + passive layer.
- ✓ Photo *fails*
- ✓ Recorded clip *fails (sequence mismatch)*
- ✓ Deepfake *must hit cues live*

IV. Methods

A monolithic auth service was rejected early: biometric ML changes weekly, but the OAuth / OIDC contract must remain frozen. The system is split along that boundary into two horizontally-scalable services on a private Docker network.

Per verification attempt:

1. Server issues a randomised action sequence $n=3-5$ and a per-attempt nonce.

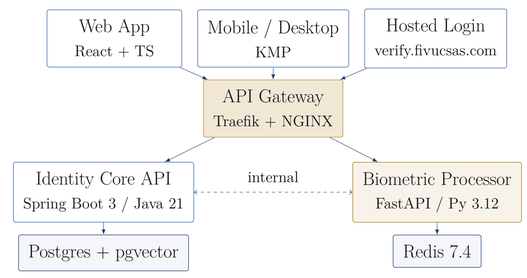
2. Client streams 60-frame landmark vectors via MediaPipe FaceLandmarker (468 points).

3. Server checks each action against EAR / MAR / yaw thresholds and temporal consistency.

4. UniFace MiniFASNet runs a passive spoof check in parallel.

5. Facenet512 embedding is encrypted (Fernet) and matched against the user vector in pgvector via an HNSW index.

V. System architecture



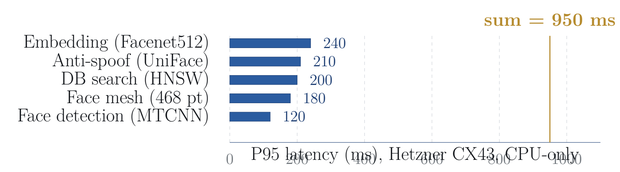
The biometric service has no public route — Docker network only, API-key gated.

VI. Authentication factors

Password · Email OTP · SMS OTP · TOTP · QR · Face · Voice · Print · WebAuthn · NFC document (MRZ + chip)

Gold = wrapped in active-liveness; any subset composes a tenant MFA flow.

VII. Experiments and results



Quality metric	Target	Observed
Replay-attack FAR ($n=120$)	$<1\,\%$	**0.0 %**
Passive-spoof AUC (internal)	>0.95	**0.97**
Backend tests	green	633 / 633
Web Vitest + Playwright	green	619 + 27
Mobile (KMP) units	green	425 / 425

VIII. Conclusion

Production-grade, GDPR-compliant biometric authentication *is* achievable on commodity hardware when the system is decomposed along its right axis — a frozen OIDC contract on one side, an iterating ML pipeline on the other — and when liveness is treated as an *active*, challenge-driven property rather than a single passive signal. Our measurements indicate that the latency budget (P95 = 950 ms on a single CPU server) and the spoof-defence profile (0 / 120 replay-attack false-accepts) are both within practical deployment limits for a tenant-shared platform, with the integrity gates (1 820+ tests, 8 ADRs, branch protection across five repositories) sufficient to keep the platform shippable through weekly ML iteration.

IX. Future work

- **BYOD tenant DB.** Enterprise tenants host biometric data in their own database, schema-pinned.
- **On-device pre-filter.** Stripped MediaPipe pipeline drops obvious non-faces before upload — bandwidth + privacy win.
- **Cross-modal challenge.** Voice nonce + facial action to defeat audio-only deepfakes.
- **Verifiable credentials.** W3C VCs after successful KYC for downstream-service portability.

X. Live demo & references

Scan for live demo
`verify.fivucsas.com`

[1] Serengil, S., Özpınar, A. *LightFace / DeepFace.* IEEE ASYU, 2020.
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[3] Soukupová, T., Čech, J. *Real-Time Eye Blink Detection using Facial Landmarks.* CVWW, 2016.
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[5] Yadav, S. et al. *UniFace: A Unified Architecture for Face Spoofing Detection.* CVPR-W, 2023.